STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: WLD-03 | TAG: NC090435

Lincoln Power MIG 255C Welder

Facility Zone: Zone 3: Welding & Thermal Processing (WLD) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. Shade 10-13 Welding Helmet + ANSI Z87.1 Safety Goggles
2. Dry Heavy Leather MIG Gloves
3. Flame-Resistant (FR) Clothing (No Open Pockets or Cuffs)
4. Fume Extraction Active

2. PRIMARY OPERATIONAL HAZARDS

1. Hot spatter, slag burns, and UV/IR arc radiation
2. Powered wire feed pinch/puncture hazard at drive rolls and gun tip
3. Thermal overload if exceeding 40% duty cycle at 250A

3. BEFORE OPERATION CHECKLIST

1. Check wire drive roll tension and liner alignment.
2. Verify DC+ polarity for solid wire with gas (or DC- for flux-core).
3. Open C25 gas cylinder slowly and set gas flow to 25-35 CFH.

4. DURING OPERATION SAFETY RULES

1. Follow the 4-inch hand clearance rule from weld zone.
2. Maintain 1/4"-3/8" wire stick-out.
3. Respect 40% duty cycle at 250A (if thermal overload trips, let cooling fan run at idle-do not cycle power).

5. AFTER USE & SHOWROOM CLEANUP

1. Shut power switch OFF, close gas bottle valve tightly, pull trigger to purge gas line, and coil leads on hangers.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Release MIG gun trigger, turn main power switch OFF, and close gas cylinder valve.